P240

3. Suppose that X is a random variable for which $E(X) = 1$, $E(X^2) = 2$, and $E(X^3) = 5$. Find the value of the third central moment of X.

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2. Suppose that X is a random variable whose distribution is completely unknown, but it is known that all the moments $E(X^k)$, for $k = 1, 2, \ldots$, are finite. Suppose also that $X_1, \ldots, X_n$ form a random sample from this distribution. Show that for $k = 1, 2, \ldots$, the kth sample moment $(1/n) \sum_{i=1}^{n} X_i^k$ is an unbiased estimator of $E(X^k)$.

3. For the conditions of Exercise 2, find an unbiased estimator of $[E(X)]^2$. *Hint:* $[E(X)]^2 = E(X^2) - \mathrm{Var}(X)$.

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6. Suppose that $X_1, \ldots, X_n$, form a random sample from a normal distribution for which the mean μ is known, but the variance σ^2 is unknown.

a. Find the moment estimate of σ^2.

b. Find the M.L.E. of σ^2.

9. Suppose that $X_1, \ldots, X_n$ form a random sample from a distribution for which the p.d.f. $f(x|\theta)$ is as follows:

$$f(x|\theta) = \begin{cases} \theta x^{\theta-1} & \text{for } 0 < x < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Also, suppose that the value of θ is unknown ($\theta > 0$).

a. Find the moment estimate of θ.

b. Find the M.L.E. of θ.

7. Suppose that $X_1, \ldots, X_n$, form a random sample from an exponential distribution for which the value of the parameter β is unknown ($\beta > 0$).

a. Find the moment estimate of β.

b. Find the M.L.E. of β.

5. Suppose that $X_1, \ldots, X_n$ form a random sample from a Poisson distribution for which the mean θ is unknown, ($\theta > 0$).

a. Determine the M.L.E. of θ, assuming that at least one of the observed values is different from 0.

b. Show that the M.L.E. of θ does not exist if every observed value is 0.

c. Show that the M.L.E. of θ is unbiased.